Lab Final

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	40
Understanding/Analysis	10					
Implementation	15					
Accuracy	10					
Task Efficiency	5					
					Total obtained mark	
Comments						

Semester: Spring2026

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Batch: 43 **Section: N2**

Course Code: SE217 **Course Name: Object Oriented Programming Lab**

Course Teacher Name: Ms. Tahmina Meem

Designation: Lecturer

Submission Date: 21/04/2026

Online Food Order & Billing System.

Code:

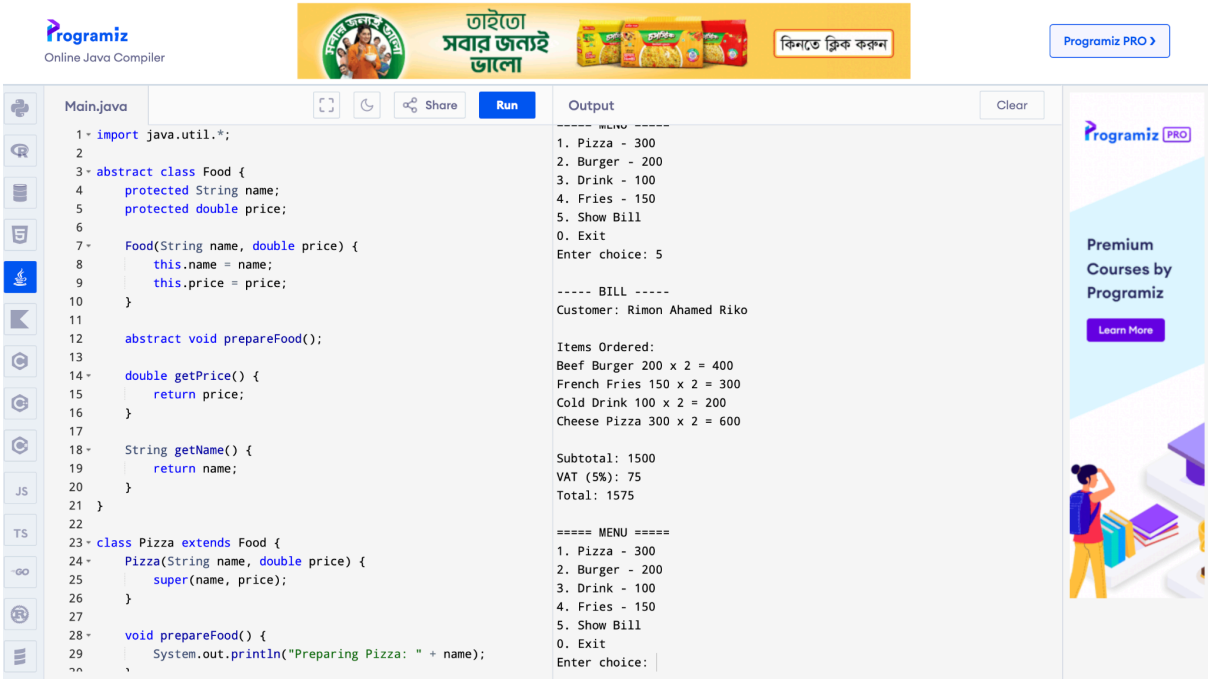
```
import java.util.*;

abstract class Food {
    protected String name;
    protected double price;

    Food(String name, double price) {
        this.name = name;
        this.price = price;
    }

    abstract void prepareFood();

    double getPrice() {
        return price;
    }

    String getName() {
        return name;
    }
}

class Pizza extends Food {
    Pizza(String name, double price) {
        super(name, price);
    }

    void prepareFood() {
        System.out.println("Preparing Pizza: " + name);
    }
}
```

```
class Burger extends Food {
    Burger(String name, double price) {
        super(name, price);
    }

    void prepareFood() {
        System.out.println("Preparing Burger: " + name);
    }
}

class Drink extends Food {
    Drink(String name, double price) {
        super(name, price);
    }

    void prepareFood() {
        System.out.println("Preparing Drink: " + name);
    }
}

class Fries extends Food {
    Fries(String name, double price) {
        super(name, price);
    }

    void prepareFood() {
        System.out.println("Preparing Fries: " + name);
    }
}
class User {
    private String name;

    User(String name) {
        this.name = name;
    }
```

```
    String getName() {
        return name;
    }
}

class Customer extends User {

    Map<String, Integer> quantityMap = new HashMap<>();
    Map<String, Double> priceMap = new HashMap<>();

    Customer(String name) {
        super(name);
    }

    void placeOrder(Food food) {
        food.prepareFood();

        String item = food.getName();
        double price = food.getPrice();

        quantityMap.put(item, quantityMap.getOrDefault(item, 0) + 1);
        priceMap.put(item, price);

        System.out.println(item + " Added. Price: " + (int)price);
    }

    void showBill() {
        double totalBill = 0;

        System.out.println("\n----- BILL -----");
        System.out.println("Customer: " + getName());

        System.out.println("\nItems Ordered:");
```

```
        for (String item : quantityMap.keySet()) {
            int qty = quantityMap.get(item);
            int price = priceMap.get(item).intValue();
            int itemTotal = price * qty;

            totalBill += itemTotal;

            System.out.println(item + " " + price + " x " + qty + " = " +
itemTotal);
        }

        int vat = (int)(totalBill * 0.05);
        int finalAmount = (int)(totalBill + vat);

        System.out.println("\nSubtotal: " + (int)totalBill);
        System.out.println("VAT (5%): " + vat);
        System.out.println("Total: " + finalAmount);
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your name: ");
        String name = sc.nextLine();

        Customer customer = new Customer(name);

        int choice;

        do {
            System.out.println("\n===== MENU =====");
            System.out.println("1. Pizza - 300");
```

```
        System.out.println("2. Burger - 200");
        System.out.println("3. Drink - 100");
        System.out.println("4. Fries - 150");
        System.out.println("5. Show Bill");
        System.out.println("0. Exit");

        System.out.print("Enter choice: ");
        choice = sc.nextInt();

        switch (choice) {
            case 1:
                customer.placeOrder(new Pizza("Cheese Pizza",
300));
                break;

            case 2:
                customer.placeOrder(new Burger("Beef Burger", 200));
                break;

            case 3:
                customer.placeOrder(new Drink("Cold Drink", 100));
                break;

            case 4:
                customer.placeOrder(new Fries("French Fries", 150));
                break;

            case 5:
                customer.showBill();
                break;

            case 0:
                System.out.println("Thank you for ordering!");
                break;
```

```
            default:
                System.out.println("Invalid choice!");
        }

    } while (choice != 0);

    sc.close();
  }
}
```

Output: